Name: _____

Question	1	2	3	4	5	*Total*
Points	6	6	6	6	6	30
Score						

1. **A prototype contest**

   Two studios compete for a contract worth 8 to Studio 1 and 6 to Studio 2. Each simultaneously chooses effort 0, 2, or 4 and pays that effort cost, whether it wins or loses. The higher effort wins. A positive tie is decided by a fair coin; if both choose 0, neither receives the contract. Payoff is expected contract value minus own effort.

   (a) (2 pts) Write the normal form, with Studio 1 choosing rows.

   (b) (2 pts) Find all pure Nash equilibria.

   (c) (2 pts) Find all Pareto optimal outcomes.

   Solution:

   (a) For a positive tie, each studio receives half its contract value before subtracting effort. The matrix is

	0	2	4
0	$0,0$	$0,4$	$0,2$
2	$6,0$	$2,1$	$-2,2$
4	$4,0$	$4,-2$	$0,-1$

   (b) Studio 1's best responses to efforts $0, 2, 4$ are, respectively, 2, 4, and $\{0, 4\}$. Studio 2's best responses to Studio 1's efforts $0, 2, 4$ are 2, 4, and 0. No cell is a mutual best response, so there is no pure Nash equilibrium.

   (c) The Pareto optimal pure profiles are $(0, 2)$, $(2, 0)$, and $(2, 2)$, with payoffs $(0, 4)$, $(6, 0)$, and $(2, 1)$. The first two give one studio its largest possible payoff. To improve on $(2, 1)$ for Studio 1 requires reducing Studio 2's payoff, and conversely. Every remaining payoff pair is dominated by one of these three.

2. **Three actions in a numerical solution**

   Consider the following game:

	L	R
U	$4,0$	$0,3$
M	$1,2$	$3,0$
D	$2,1$	$2,1$

(a) (3 pts) Find a Nash equilibrium in which Row uses all three actions and assigns probability $\frac{1}{2}$ to D. Verify it and give both payoffs.

(b) (3 pts) Column plays pure strategy R. For $e^{\lambda} = 2$, compute Row's logit response. Does pairing this response with pure R give a logit QRE at this finite λ? Explain.

Solution:

(a) Against uniform Column play, all three Row actions give 2, so Row's current payoff is 2 and $\delta_1 = 0$. Column's pure payoffs are

$$U_2(p_1, L) = 1, \qquad U_2(p_1, R) = \tfrac{5}{4}.$$

Its current payoff is $\frac{1}{2}(1 + \frac{5}{4}) = \frac{9}{8}$, hence $\delta_2 = \frac{5}{4} - \frac{9}{8} = \frac{1}{8}$. Thus $\epsilon_{\min} = \frac{1}{8}$.

(b) Write $q = P(L)$. Row's three pure payoffs are $4q$, $3 - 2q$, and 2. All are equal at $q = \frac{1}{2}$. If Row's probabilities on U, M are a, b, then $a + b = \frac{1}{2}$. Column's indifference condition is

$$2b + \tfrac{1}{2} = 3a + \tfrac{1}{2}.$$

Consequently $a = \frac{1}{5}$, $b = \frac{3}{10}$. The equilibrium is

$$p_1 = (\tfrac{1}{5}, \tfrac{3}{10}, \tfrac{1}{2}), \qquad p_2 = (\tfrac{1}{2}, \tfrac{1}{2}), \qquad (U_1, U_2) = (2, \tfrac{11}{10}).$$

Every pure action of each player gives its equilibrium payoff, so no unilateral deviation is profitable.

(c) Against R, Row's payoffs are $0, 3, 2$, giving logit weights $1, 8, 4$. Its response is therefore $(\frac{1}{13}, \frac{8}{13}, \frac{4}{13})$. The resulting profile is not a logit QRE: at finite precision, Column's logit response assigns positive probability to both actions and cannot be pure R. Both players must satisfy the response condition.

3. **Certifying a security strategy**

Consider the zero-sum game:

	L	C	R
U	0	3	2
M	4	0	4
D	4	2	0

(a) (1 pt) Find the maximin and minimax values in pure strategies. Is there a pure Nash equilibrium?

(b) (2 pts) Write Row's linear program for an optimal mixed strategy. Why is it enough to check it against the pure strategies of Column only?

(c) (2 pts) Row proposes $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. Find a Column strategy that shows its optimality, and determine the game's value.

(d) (1 pt) Which pure profiles are Pareto optimal? Justify without comparing all pairs of cells individually.

Solution:

(a) All row minima are 0, so the pure maximin value is 0. Column maxima are $4, 3, 4$, so the pure minimax value is 3. There is no pure saddle point.

(b) Let x_U, x_M, x_D be Row's probabilities. Its LP is

$$\begin{aligned} \max \quad & v \\ \text{s.t.} \quad & 4x_M + 4x_D \geq v, \\ & 3x_U + 2x_D \geq v, \\ & 2x_U + 4x_M \geq v, \\ & x_U + x_M + x_D = 1, \quad x_U, x_M, x_D \geq 0, \quad v \in \mathbb{R}. \end{aligned}$$

Each inequality guarantees v against one pure column. Against a mixture, the payoff is an average of these three payoffs, still at least v.

(c) The proposed Row strategy gives 2 against every column, so it guarantees 2. To hold each row to 2, seek probabilities y_L, y_C, y_R with

$$3y_C + 2y_R = 2, \quad 4y_L + 4y_R = 2, \quad 4y_L + 2y_C = 2.$$

Together with normalization these give $y = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$. Each pure row gives exactly 2 against this mixture. Hence Column holds Row to at most 2; both strategies are optimal and $v = 2$.

(d) All nine pure profiles are Pareto optimal. Total payoff is always zero: increasing one player's payoff necessarily decreases the other's. Thus no profile Pareto dominates another.

4. **Private recommendations with three row actions**

A device draws a pair of actions from the distribution p below and privately recommends each player only their own action. The game and recommendation rule are publicly known.

	L	R
U	$4, 2$	$0, 3$
M	$0, 1$	$4, 4$
D	$3, 3$	$3, 1$

p	L	R
U	$\frac{1}{4}$	0
M	0	$\frac{1}{4}$
D	$\frac{1}{4}$	$\frac{1}{4}$

(a) (3 pts) Is p a correlated equilibrium? Check every recommendation that occurs, allowing deviations to every other action.

(b) (3 pts) Replace p by independent draws from its two marginal distributions. Is the new distribution a CE? If not, exhibit a profitable deviation conditional on a recommendation and calculate its gain.

Solution:

(a) Expected payoffs are

$$U_1 = \tfrac{1}{4}(4+4+3+3) = \tfrac{7}{2}, \qquad U_2 = \tfrac{1}{4}(2+4+3+1) = \tfrac{5}{2}.$$

Row is told D with probability $\frac{1}{2}$, so conditional on D, Column's recommendations L, R each have probability $\frac{1}{2}$.

(b) Yes. Row's conditional comparisons are:

- Told U: Column plays L; obeying gives 4, versus 0 from M and 3 from D.
- Told M: Column plays R; obeying gives 4, versus 0 from U and 3 from D.
- Told D: Column is uniform; obeying gives 3, versus 2 from either U or M.

Column told L faces U, D equally: obeying gives $\frac{1}{2}(2+3) = \frac{5}{2}$, whereas R gives $\frac{1}{2}(3+1) = 2$. Column told R faces M, D equally: obeying gives $\frac{1}{2}(4+1) = \frac{5}{2}$, whereas L gives $\frac{1}{2}(1+3) = 2$. Every obedience comparison holds strictly.

(c) The marginals are $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ and $(\frac{1}{2}, \frac{1}{2})$. With independent draws, Row told U still faces uniform Column play. Obeying gives 2 while switching to D gives 3, a conditional gain of 1. The product distribution is not a CE. The original recommendation conveys information about the other action; independent draws remove that information.

5. **A commitment with three available actions**

Row publicly commits to probabilities (x, y, z) on U, M, D. Column observes these probabilities, not the realized row, and best responds.

	L	R
U	$3, 3$	$2, 1$
M	$1, 2$	$4, 2$
D	$6, 1$	$1, 3$

(a) (4 pts) Write and solve the two leader LPs, one per follower action. Find the strong Stackelberg equilibrium and both payoffs. Explain meaning of all parts of the linear program. Hint: besides feasible pure strategies, the only vertex of either feasible polygon is $(\frac{1}{2}, 0, \frac{1}{2})$.

(b) (2 pts) Does a weak Stackelberg equilibrium exist? If not, give the supremum and commitments approaching it.

Solution:

(a) Column's payoffs are $3x+2y+z$ from L and $x+2y+3z$ from R. Thus L is best when $x \geq z$ and R when $z \geq x$; there is a tie exactly when $x = z$.

(b) Both LPs impose $x+y+z=1$ and $x,y,z \geq 0$:

$$\begin{aligned} LP_L: &\quad \max(3x+y+6z) \quad \text{s.t. } x \geq z, \\ LP_R: &\quad \max(2x+4y+z) \quad \text{s.t. } z \geq x. \end{aligned}$$

For LP_L, vertices $U, M, \frac{1}{2}U+\frac{1}{2}D$ give values $3, 1, \frac{9}{2}$. For LP_R, vertices $D, M, \frac{1}{2}U+\frac{1}{2}D$ give $1, 4, \frac{3}{2}$. The SSE is therefore $(x,y,z) = (\frac{1}{2}, 0, \frac{1}{2})$ with response L, giving payoffs $(\frac{9}{2}, 2)$. At this commitment, strong tie-breaking selects L because Row receives $\frac{9}{2}$ rather than $\frac{3}{2}$.

(c) No WSE exists. The unique maximizer of LP_L is the commitment in (b), but weak tie-breaking selects R there, giving Row $\frac{3}{2}$. Every other commitment inducing L gives less than $\frac{9}{2}$, and every response R gives at most 4. For $0 < \eta < \frac{1}{2}$, the commitment

$$(x,y,z) = (\tfrac{1}{2}+\eta, 0, \tfrac{1}{2}-\eta)$$

induces L uniquely and gives Row $\frac{9}{2}-3\eta$. Thus the weak supremum is $\frac{9}{2}$, approached as $\eta \downarrow 0$ but never attained.